Exercise #3: Application Programming Interface (API) – Based Integration

API-Based System Integration

- ✓ An API is a set of rules, protocols, and tools that allow different software applications to communicate with each other. It enables direct interaction between applications, systems, or services.
- ✓ Acts as a bridge between applications.
- ✓ API-based integration allows different systems (e.g., your university web system and mobile app) to communicate using web services.

Learning Objective:

Integrate a mobile-based student interface using Android Studio and Java with the web-based University Management System via API. Specifically, we aim to enable university students to:

1. **enroll course**, and **view grades using android mobile app**
2. **develop the "Management Information System (MIS)" web interface**
3. **connect MIS with the University System database**

Part 1. Setting Up the Web API (PHP & MySQL)

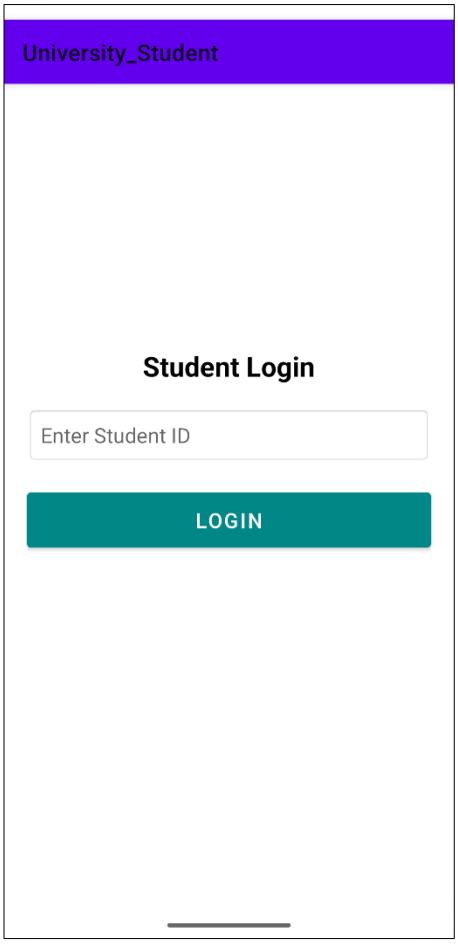
Part 2: Students Mobile Application development using Android Studio

Part 3 : Set Up a Student Mobile App in Android Studio with MySQL and Apache Using API

Part 4. Web-based "Management Information System" development

Part 3 : Set Up a Student Mobile App in Android Studio with MySQL and Apache Using API

1. **Student Login**
   - ☐ First, we need to set **LoginActivity** as the initial screen when the app launches. This ensures that only students whose **records are stored in the database or server** (Apache MySQL) can access the mobile app by logging in with their credentials.
     - ✓ Update your **AndroidManifest.xml** file and set **LoginActivity** as the main entry point.




```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.university_system"
    xmlns:tools="http://schemas.android.com/tools">

    <!-- 📌 Add necessary permissions -->
    <uses-permission android:name="android.permission.INTERNET" />
    <uses-permission
android:name="android.permission.ACCESS_NETWORK_STATE" />

    <application
        android:allowBackup="true"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.UniversityStudent"
        tools:targetApi="31">

        <!-- 🔑 Login Screen (Now the Launcher) -->
        <activity
            android:name=".ui.LoginActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>

        <!-- 🚀 Main Dashboard -->
        <activity android:name=".ui.MainActivity" android:exported="true" />

        <!-- 📝 Enrollment Screen -->
        <activity android:name=".ui.EnrollActivity" android:exported="false" />

        <!-- 📊 View Grades Screen -->
        <activity android:name=".ui.ViewGradesActivity"
android:exported="false" />

    </application>

</manifest>
```

✓ Inside the **XAMPP's htdocs > university_api** folder, create a **login.php** file to check student login credentials

```
<?php
header("Content-Type: application/json");
header("Access-Control-Allow-Origin: *");
header("Access-Control-Allow-Methods: POST, GET");
header("Access-Control-Allow-Headers: Content-Type");

// ✅ Database Connection
$conn = new mysqli("localhost", "root", "", "university_db");

// ✅ Check connection
if ($conn->connect_error) {
    die(json_encode(["error" => "Database connection failed: " . $conn->connect_error]));
}

// ✅ Get JSON data from request (for POST requests)
$data = json_decode(file_get_contents("php://input"), true);

// ✅ Check both GET and POST for student_id
$student_id = isset($data['student_id']) ? $data['student_id'] : (isset($_GET['student_id']) ?
$_GET['student_id'] : null);

if (!$student_id) {
    echo json_encode(["error" => "Missing student_id"]);
    exit();
}

// ✅ Prepare SQL query to fetch student details securely
$stmt = $conn->prepare("SELECT student_id, CONCAT(first_name, ' ', last_name) AS full_name, program
FROM students WHERE student_id = ?");
$stmt->bind_param("i", $student_id);
$stmt->execute();
$result = $stmt->get_result();

if ($result->num_rows > 0) {
    $row = $result->fetch_assoc();

    // ✅ Check if `full_name` is NULL
    if (empty($row["full_name"])) {
        echo json_encode(["error" => "Student record incomplete. Full name missing."]);
        exit();
    }

    // ✅ Return student details
    echo json_encode([
        "status" => "success",
        "student_id" => $row["student_id"],
        "name" => $row["full_name"],
        "program" => $row["program"]
    ]);
} else {
    echo json_encode(["error" => "Student not found"]);
}

// ✅ Close connections
$stmt->close();
$conn->close();
?>
```

✓ Test your API using Postman:

1. Open **Postman** and send a **POST** request to:

```
http://localhost/university_api/login.php?student_id=2
```

2. Add a parameter:
   - Key: **student_id**
   - Value: **2**

3. Click **Send**.
4. If student 2 exists in MySQL, it will return:



✓ **Integrate the API with Your Android App**

Now that the API works, let's modify your Android app to fetch student credentials from MySQL and log in using student_id.

1. Update **LoginActivity.kt** to **Call the API** and send **student_id**

```
package com.example.university_system.ui

import android.content.Intent
import android.os.Bundle
import android.widget.Button
import android.widget.EditText
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity
import com.android.volley.Request
import com.android.volley.Response
import com.android.volley.toolbox.JsonObjectRequest
import com.android.volley.toolbox.Volley
import com.example.university_system.R
import org.json.JSONObject

class LoginActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_login)

        val studentIdInput = findViewById<EditText>(R.id.et_student_id)
        val loginButton = findViewById<Button>(R.id.btn_login)
```

```
        loginButton.setOnClickListener {
            val studentId = studentIdInput.text.toString().trim()

            if (studentId.isEmpty()) {
                Toast.makeText(this, "Please enter your Student ID", Toast.LENGTH_SHORT).show()
            } else if (!studentId.matches(Regex("\\d{5,10}"))) { // Ensure Student ID is numeric (5-10
digits)
                Toast.makeText(this, "Invalid Student ID. Must be 5-10 digits.",
Toast.LENGTH_SHORT).show()
            } else {
                loginUser(studentId)
            }
        }
    }

    private fun loginUser(studentId: String) {
        val url = "http://10.0.2.2/university_api/login.php" // ✅ Use localhost for Android Emulator

        val requestQueue = Volley.newRequestQueue(this)

        // ✅ Prepare JSON request body
        val jsonRequest = JSONObject()
        jsonRequest.put("student_id", studentId)

        // ✅ Create API request
        val request = JsonObjectRequest(
            Request.Method.POST, url, jsonRequest,
            Response.Listener { response ->
                if (response.has("status") && response.getString("status") == "success") {
                    val name = response.getString("name")
                    val program = response.getString("program")

                    Toast.makeText(this, "Welcome, $name!", Toast.LENGTH_SHORT).show()

                    // ✅ Redirect to MainActivity with student details
                    val intent = Intent(this, MainActivity::class.java)
                    intent.putExtra("STUDENT_ID", studentId)
                    intent.putExtra("STUDENT_NAME", name)
                    intent.putExtra("STUDENT_PROGRAM", program)
                    startActivity(intent)
                    finish()
                } else {
                    val errorMessage = response.optString("error", "Login failed")
                    Toast.makeText(this, errorMessage, Toast.LENGTH_SHORT).show()
                }
            },
            Response.ErrorListener { error ->
                Toast.makeText(this, "Login Error: ${error.message}", Toast.LENGTH_SHORT).show()
            }
        )

        // ✅ Add request to queue
        requestQueue.add(request)
    }
}
```

- **activity_login.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:gravity="center"
    android:background="@android:color/white">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Student Login"
        android:textSize="24sp"
        android:textStyle="bold"
        android:textColor="@color/black"
        android:gravity="center"
        android:layout_marginBottom="20dp" />

    <EditText
        android:id="@+id/et_student_id"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Student ID"
        android:inputType="number"
        android:maxLength="10"
        android:padding="12dp"
        android:background="@android:drawable/editbox_background"
        android:textSize="18sp"
        android:layout_marginBottom="20dp"/>

    <Button
        android:id="@+id/btn_login"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Login"
        android:backgroundTint="@color/teal_700"
        android:textColor="@android:color/white"
        android:textSize="18sp"
        android:padding="12dp"/>

</LinearLayout>
```

✓ Modify **MainActivity.kt** to **receive and display student details**.

```
package com.example.university_system.ui
import com.example.university_system.R
import android.content.Intent
import android.os.Bundle
import android.widget.Button
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

   override fun onCreate(savedInstanceState: Bundle?) {
      super.onCreate(savedInstanceState)
      setContentView(R.layout.activity_main)

      // Fetch UI elements
      val tvWelcomeMessage: TextView = findViewById(R.id.tvWelcomeMessage)
      val tvStudentId: TextView = findViewById(R.id.tvStudentId)
      val tvStudentName: TextView = findViewById(R.id.tvStudentName)
      val tvStudentProgram: TextView = findViewById(R.id.tvStudentProgram)
      val btnEnrollCourses: Button = findViewById(R.id.btnEnrollCourses)
      val btnViewGrades: Button = findViewById(R.id.btnViewGrades)
      val btnLogout: Button = findViewById(R.id.btnLogout)

      // Get student details from intent (assuming they are passed from LoginActivity)
      val studentId = intent.getStringExtra("STUDENT_ID") ?: "N/A"
      val studentName = intent.getStringExtra("STUDENT_NAME") ?: "N/A"
      val studentProgram = intent.getStringExtra("STUDENT_PROGRAM") ?: "N/A"

      // Set the student details
      tvWelcomeMessage.text = "Welcome, $studentName!"
      tvStudentId.text = "Student ID: $studentId"
      tvStudentName.text = "Name: $studentName"
      tvStudentProgram.text = "Program: $studentProgram"

      // Button to navigate to EnrollCoursesActivity
      btnEnrollCourses.setOnClickListener {
         val intent = Intent(this, EnrollActivity::class.java)
         intent.putExtra("STUDENT_ID", studentId)
         startActivity(intent)
      }

      // Button to navigate to ViewGradesActivity
      btnViewGrades.setOnClickListener {
         val intent = Intent(this, ViewGradesActivity::class.java)
         intent.putExtra("STUDENT_ID", studentId)
         startActivity(intent)
      }

      // Logout button: Redirects to LoginActivity
      btnLogout.setOnClickListener {
         val intent = Intent(this, LoginActivity::class.java)
         intent.flags = Intent.FLAG_ACTIVITY_NEW_TASK or
Intent.FLAG_ACTIVITY_CLEAR_TASK
         startActivity(intent)
         finish()
      }
   }
}
```

✓ Modify your UI to **include a welcome message** and a **logout button**.
activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp"
    android:background="@color/white">

    <!-- Welcome Message -->
    <TextView
        android:id="@+id/tvWelcomeMessage"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome, Student!"
        android:textSize="22sp"
        android:textStyle="bold"
        android:padding="16dp"
        android:gravity="center"
        android:textColor="@color/black" />

    <!-- Student Information -->
    <TextView
        android:id="@+id/tvStudentId"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Student ID: "
        android:textSize="18sp"
        android:textColor="@color/black"
        android:padding="4dp" />

    <TextView
        android:id="@+id/tvStudentName"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Name: "
        android:textSize="18sp"
        android:textColor="@color/black"
        android:padding="4dp" />

    <TextView
        android:id="@+id/tvStudentProgram"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Program: "
        android:textSize="18sp"
        android:textColor="@color/black"
        android:padding="4dp" />

    <!-- Enroll in Courses Button -->
    <Button
        android:id="@+id/btnEnrollCourses"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Enroll in Courses"
        android:textSize="16sp"
        android:backgroundTint="@color/teal_700"
```

```
        android:textColor="@color/white"
        android:padding="12dp"
        android:layout_marginTop="24dp" />

    <!-- View Grades Button -->
    <Button
        android:id="@+id/btnViewGrades"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="View Grades"
        android:textSize="16sp"
        android:backgroundTint="@color/teal_700"
        android:textColor="@color/white"
        android:padding="12dp"
        android:layout_marginTop="12dp" />

    <!-- Logout Button -->
    <Button
        android:id="@+id/btnLogout"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Logout"
        android:textSize="16sp"
        android:backgroundTint="@color/red"
        android:textColor="@color/white"
        android:padding="12dp"
        android:layout_marginTop="12dp" />

</LinearLayout>
```

- Update **AndroidManifest.xml** to allow Cleartext Traffic (Fix for HTTP Error)

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.university_system"
    xmlns:tools="http://schemas.android.com/tools">

    <!-- 📌 Add necessary permissions -->
    <uses-permission android:name="android.permission.INTERNET" />
    <uses-permission android:name="android.permission.ACCESS_NETWORK_STATE" />

    <application
        android:usesCleartextTraffic="true"
        android:allowBackup="true"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.UniversityStudent"
        tools:targetApi="31">

        <!-- 🔑 Login Screen (Now the Launcher) -->
        <activity
            android:name=".ui.LoginActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
```

```
        </activity>

        <!-- 🚀 Main Dashboard -->
        <activity android:name=".ui.MainActivity" android:exported="true" />

        <!-- 📝 Enrollment Screen -->
        <activity android:name=".ui.EnrollActivity" android:exported="false" />

        <!-- 📊 View Grades Screen -->
        <activity android:name=".ui.ViewGradesActivity" android:exported="false" />

    </application>

</manifest>
```

- Updated **colors.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="purple_200">#FFBB86FC</color>
    <color name="purple_500">#FF6200EE</color>
    <color name="purple_700">#FF3700B3</color>
    <color name="teal_200">#FF03DAC5</color>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFFF</color>
    <color name="red">#D32F2F</color> <!-- Example Red -->
    <color name="teal_700">#00796B</color> <!-- Example Button Color -->
</resources>
```

- Updated **styles.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <style name="Theme.UniversityStudent"
parent="Theme.MaterialComponents.Light">
        <!-- Your theme settings -->
    </style>
</resources>
```

Part 3: Final Output

